The responsive museum

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Abstract: This paper reports on the ongoing work carried out by V-Must, a Network of Excellence, funded by the European FP in its efforts to rethink the virtual museum (VM). The Network has been active in identifying, and mapping tools, and services that define and support VMs in the heritage sector. V-MUST.NET is coordinated by CNR and includes 18 Partners from 13 countries, and Associated Members. The Network was launched 1st of February 2011 and continues through to 31st of January 2015.

Drawing on a series of reports and publications prepared by the Network, http://www.v-must.net/library/publications, we will continue to reflect on the VM from a variety of perspectives.

This paper looks at the responsive screen, dynamically re-sizing itself for numerous platforms, including: in-house large screens, PC, mobile, and tablet. We also critically consider the response of the user/visitor who will be encountering the Museum before, and after the visit as well as during the actual visit. What kinds of implications will this have on the visit and visitor, and how can the Museum prepare for these different kinds of scenarios?

Through a series of three case studies we attempt to define the VM as we re-visit the core concept of the museum ethos as it reaches out to meet its visitor. Introducing the visitor to the exhibition, even before the physical visit in an exceptionally well-honed marketing scenario, we will discuss the virtual bear hug, where the Arthur M. Sackler Gallery in Washington wraps its electronic arms around the future visitor to the exhibition Yoga: The Art of Transformation. The deluge of visitor/museum scenarios, including promoting the show over numerous social networks and enticing invitations to 'use' the exhibition in novel ways aims to introduce the visitors to the exhibition which opened October 19, 2013 - well before the red ribbon was cut. The second case study describes an engaging scenario of electronic delivery that accompanies the visitor during the visit and describes the CMA Collection Wall, a 40-foot multi-touch MicroTile Collection Wall that dramatically visualizes all the works currently on view in CMA's permanent collection galleries, plus some that are in storage, totaling over 3,800 works of art. The third VM scenario, Ask Jacques Lipchitz a Question, authored by the Israel Museum, Jerusalem can be enjoyed after the visit and serves, in a novel way, to augment and enhance the experience and maintain the connection between the Museum and the visitor opening up opportunities to 'meet the artist' and 'hear his voice', even after his own death.

Keywords: responsive design; ecomuseum, virtual museum

Introduction

The expression 'responsive design' is a term that we are hearing about more and more, but how does this relate to websites and in what ways can this concept possibly be connected to museums? Architects and engineers are experimenting with motion sensors that respond to the presence of the people moving within the environment; adjusting, for example, the room's temperature and triggering pre-synched systems to prompt ambient lighting. This approach to physical spaces is now referred to as responsive architecture[1] where embedded systems sense presence, and motion and prompt the environment to adjust in return. Physical spaces essentially enter into conversation with the people who occupy them, and respond in real time, accommodating them accordingly.

In a networked world, other kinds of systems need to develop different responsive solutions to deliver rich content to a range of platforms: large screens, pc's, mobile phones, tablets, etc... These systems also react in real time. Applying the same kind of responsive principle to optimize the viewing experience – easy reading and navigation with a minimum of re-sizing, panning, and scrolling – across a wide range of screen-sizes (from desktop computer monitors, to tablets, to mobile phones), this approach assures efficient delivery of content to all users, whatever the choice of platform.

This paper takes this approach one step further and explores what kinds of conversations are possible once the museum engages with audiences. Enabling access to exhibitions, collections, events and educational activities, it essentially facilitates public conversation around issues that concern the Museum. The responsive design approach, therefore, when applied to the Museum, serves to foster dialogue between the Museum and its visitors, inspiring the public to join in the exchange and encouraging truly reciprocal conversations. This paper discusses how the virtual museum (VM) opens up new possibilities to harness, and to enact reciprocal, user-driven scenarios, and furnishes new opportunities for the remote visitor to be able to interact with the physical museum in novel ways.

[1] <http://en.wikipedia.org/wiki/Responsive_architecture>

To explore the VM in the context of the responsive museum, we also reflect on the practice of new museology, not as a specific turning point in the history of museums, but rather as a marker of the on-going, re-evaluation of the museum in relationship with its audiences.[2] New museology has at times implied a radical re-organisation of museum agendas, such as a move from an elitist, undemocratic space towards a more democratic space, the prioritising of the visitor rather than the object, or the reclaiming, or re-territorising of the museum as a space that could be owned by the community.[3] At the same time, a similar expectation of 'newness' is implicit in the term 'new media' when applied in the context of the museum. However, both terms have been repeated so often that they are at risk of becoming redundant and need to be re-appraised before applying them to the responsive museum. While the move from 'old' to 'new' media describes a modification of the technological platform, rather than a radical change in content **–** from traditional print distribution and terrestrial television, to webcasting, podcasting and electronic peer to peer communications over the Internet **–** the term 'new museology' is invoked to suggest changes in institutional ideologies. While this paper uses both of the terms to suggest a break from the corresponding 'old' agendas, it also questions whether the idea of 'newness' inherent in both of these expressions simply represents a modification of the platform of delivery, rather than a radicalisation of content, ideology or institutional agendas. We argue that new media iterations of museum agendas, in fact, often deliver applications that represent electronic versions of the old practices, and, with these applications the underlying ideologies of the inherited legacies are replicated and consequently re-distributed. The use of the terms that invoke the illusion of 'newness', therefore, may set up false expectations, especially when electronic environments are perceived as a panacea for archaic and inefficient systems.

In a presentation on e-government and the Information Society, Paul Timmers, Head of the Unit for E-Government in the European Commission, Directorate-General Information Society, argued that, while those who develop these initiatives, as well as those who will be using them, expect innovation, the emergence of new skills, and organizational implementation through the uptake of ICTs, are often based on, and essentially replicate, the old systems with all their inherent problems.[4] According to Timmers, in reproducing legacy systems, albeit in an electronic configuration, environments that articulate legislative practices **-** such as paying fines or taxes online or renewing a passport or driver's licence - that were cumbersome and outdated in their underlying assumptions, are often driven by the same organizational principles when transposed online.

The ecomuseum, new museology, and cultural diversity

Both the national and the universal museum have their own stories to tell, and each prioritises their collections and exhibitions accordingly with these agendas in mind. The theory and practice of the new museologists built on the ecomuseum, which emphasised a community–driven agenda, where the public was encouraged to take on an active role, and where a plurality of voices, rather than the single authoritarian voice could be heard. This section introduces new museology and its prioritising of cultural diversity, which, in contrast to the universal or national museum, actively encourages visitor participation, and visibly welcomes contributions from the community.

The museum is located within the reach of the community, and as each museum conceptualises 'its community' in specific ways these different perspectives inevitably determine the kinds of visits the public will experience when they visit. Whether the visit is purely educational or for pleasure, visitors will encounter the institutional narrative, and whether the visitor complies or not with this narrative, determines if he or she feels included or excluded from the experience. Describing the tools and processes by which concepts such as the 'museum' become self-legitimating, Irit Rogoff (1994, p.232) argues: 'the museum as a complex amalgam of ideological intentions operating through strategies of pleasure and gratifications, is equally the site of the production of cultural identities. Through numerous and varied practices cultural exclusions are reproduced and cultural "otherness" is constituted'. The display of meta-narratives and micro-narratives of cultural heritage in the gallery make the divisions of sub-cultures and para-cultures highly visible, therefore legitimising some kinds of cultural affiliation, while presenting other kinds of cultural practice as exotic or alien. When 'normative' displays of 'our' heritage, or 'our' shared memory, refer only to some members of society, not all

[2] According to Patrick Boylan, the term 'new museology' was first introduced in the United States in 1958 by Mills and Grove in their contribution to S. De Borghegyi's book *The Modern Museum and the Community*. From compilation of museum definitions: Patrick Boylan, City University. <xa.yimg.com/kq/groups/.../Defining+Museums+and+Galleries.doc>.

[3] See for example the various projects and agendas described in Hauenschild (1988).

[4] The symposium was held at the Internet Committee Meeting, Israel Knesset (Parliament), Jerusalem, January 18, 2005.

visitors will necessarily concur with this message. Visitors who comply with the story line will readily engage with the narrative; alternatively, the exhibition message may be read in opposition, while at other times visitors may simply react indifferently.

While exclusion is often (although not exclusively) constructed as a derogatory or subordinate position, museum narratives often set up binary opposites to create cultural specificity, whether they are historical, ethnic, or geographical.[5] Karp (1991, p.375) referred to 'exoticizing' and 'assimilating' strategies that produce different kinds of responses to describe the different options open to the museum; these micro-narratives come together to construct the meta-narrative that visitors encounter when they come into the museum. As visitors enter the gallery, they do so with their own pre-figured, cultural empathies and alliances. See for example Doering's (1999, p.8) discussion on *entrance narratives*, or the *internal story line*. At times, the visitors embrace the narrative, while at others times the narrative is read against the grain, resulting in alienation and a sense of cultural exclusion.[6] Much effort goes into promoting social inclusion in the museum (see for example, the Group for Large Local Authority Museums, GLLAM Report, 2000, and the DCMS Department of Culture, Media and Science toolkit), and many museums' educational activities are now sensitive to the possibilities that some people may feel excluded.[7] The examples in this paper look at the narratives produced by the museum of the *new* museologists, those museums that choose to create true conversations through new media. We consider the ways that these kinds of museums produce community-driven scenarios and explore how these VMs mediate their messages to act as a site of production of cultural identity, with an agenda that promotes social inclusion.

While we are clearly discussing the kinds of experiences that are aligned to the Web 2.0 Museum, a term coined by Nino Simon in her popular blog and seminal publication[8], the historical process of the new museologist conceptually prefaced the digital interaction of a community driven museum. The ecomuseum was introduced in the late 1970s and early 80s in France in a bid to reaffirm a sense of community through critique of the object-orientated model of the museum. According to Poulet (1994, p.71), the term was forged in 1971, during the ninth meeting of ICOM in Grenoble, when 'the idea of heritage [*patrimoine*] linked to specific communities and localities had begun to interest the French Ministry of the Environment and high officials in charge of the nation's physical resources'. The ecomuseum, Poulet argues, 'spawned a distinctive conception of identity, according to which the preservation of culture was a kind of social responsibility' (ibid.). The community of Le Creusot-Montceau-les-Mines, Poulet explains, was created in 1974 by Marcel Evard in collaboration with Rivière. Four distinct fields of activity were envisioned by the founders: remembrance; understanding [*la connaissance*]; a joint management and development of the locale by the inhabitants and a team of scientists; and finally, artistic creation (arts retreats; creative arts projects linked to local industry and technology) (1994, pp.71-72). Community-orientation and the active participation by the community as actors and even authors of the shared projects drove these kinds of moves. When inaugurating the Fresnes Ecomuseum in 1978, in a suburb of Paris, Georges Henri Rivière, the first Director of ICOM, outlined the underlying principles by which a community could view itself as 'an ecomuseum,' describing the ecomuseum as:

> An instrument conceived, fashioned and operated jointly by a public authority and a local population. It is a mirror in which the local population views itself to discover its own image, in which it seeks an explanation of the territory to which it is attached and of the populations which have preceded it, through the discontinuity or continuity of generations. (Rivière 1978, quoted in Delarge 2001)[9]

The ecomuseum encouraged local cooperation and pro-active contributions of the community that determined exhibition strategies and educational agendas. For example, collaboration between Fresnes town and the community included several projects: a celebration that marked the local

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[5] See, for example, Frantz Fanon's (1967) description of how the category "white" is dependent for its stability on its negation of "black" where Fanon's binary Self/Other of the colonized and colonizer is projected onto the racially determined category of Black.

[6] See Comolli and Narboni on reading against the grain in cinematic reading (1977), and Stuart Hall's model of mass communication which stressed active interpretation within the relevant codes of encoding and decoding (1999, pp.123-38).

[7] Libraries, Museums, Galleries and Archives for All: Co-operating Across the Sectors to Tackle Social Exclusion, January 2001, http://books.google.co.il/books/about/Libraries_Museums_Galleries_and_Archives.html?id=3zKNYgEACAAJ&redir_esc=y

[8] http://museumtwo.blogspot

[9] ICOM News, Volume 54 - 2001 N°1 <http://www.chin.gc.ca/Applications_URL/icom/managing_change.html>.

prison's centenary, and the presentation of *Rassemblance*, an exhibition that dealt with immigration. *Rassemblance* offered an opportunity for members of the immigrant population of Fresnes to portray one hundred years of immigration to France, gathering the photographic information that illustrated their own stories and their own heritage from their 'home' surroundings (ibid.).

The idea of 'a new museology' arrived in the UK at the end of the 1980s, soon after the ecomuseum had appeared in France. In the UK at this time, the museum community faced a crisis: funding was becoming scarce and museums were concerned with losing their direction.[10] The publication, *The New Museology*, edited by Peter Vergo (1989), included nine authors who introduced a series of radical challenges to the *old* view of a museum, calling for a reform of traditional museology, with a move towards community action. Andrea Witcomb (2003), an Australian-based, self-declared, new museologist describes 'New Museology', explaining that 'one of the ways in which contemporary museums attempt to challenge dominant views of the museum as site of power relations is to invoke and encourage new relations between museum and communities'. Witcomb suggests resolving the tensions through innovative curatorial practice, so that 'one way of avoiding romantic notions of community, while also recognizing that museums are engaged in dialogue, would be to think of museums *themselves* as communities' [original italics] (2003, p.79). She argues that this is demonstrated through her own curatorial practice at the Fremantle History Museum in Western Australia in a case study, *Travelers and Immigrants* (ibid.: 86). As a curator, Witcomb declares that she wanted the exhibition 'to be attentive to the problem of "voice" […] and to reflect the meanings Portuguese-Australian people themselves gave to the objects as symbols of their own migrant experience'.

Attention to voice and authorship that depicts and structures community narratives could now be reshuffled in the light of emerging technologies, where VMs could present opportunities for voices to emerge from grass roots communities, perhaps for the first time. One of the first expressions of these kinds of platforms was *Moving Here*, the online stories of Caribbean, Irish, Jewish and South Asian people who left their homelands to move to England over the previous 200 years. This was one of the earliest examples of a new museological approach that was attentive to voices from the community and supported by new media, enabling new kinds of synergies to develop.[11] The online interface not only disseminated the archived histories from 30 museums across the UK in novel ways, but also offered opportunities for remote visitors to contribute their own micro-histories to the London community and their shared memory. Reaching out to the community in this way had its own historical antecedents, and represented a new chapter in the continuing evolution of the modern museum.

Clifford has described museums as 'contact zones', as discursive spaces where 'aspirations of both dominant and subaltern populations can be articulated through this structure, along with the material interests of nation and national tourism' (Clifford 1997, p.218). Clifford borrows the term 'contact zones' from Mary Louise Pratt, who in her book *Imperial Eyes: Travel and Transculturation,* defines the 'contact zone' as 'the space of colonial encounters, the space in which peoples geographically and historically separated come into contact with each other and establish ongoing relations, usually involving conditions of coercion, radical inequality, and intractable conflict' (Pratt 1992, pp.6-7, quoted in Clifford 1997, p.192).

For Clifford, however, the idea of a contact zone

> can be extended to include cultural relations within the same state, region, or city – in the centers rather than the frontiers of nations and empires. The distances at issue here are more social than geographic. For most inhabitants of a poor neighborhood, located perhaps just blocks or a short bus ride from a fine-arts museum, the museum might as well be on another continent. (Clifford 1997, p.204)

Clifford's account of museums as contact zones 'argues for a democratic politic that would challenge the hierarchical valuing of different places of crossing. It argues for a decentralization and circulation of collections in a multiple public sphere, an expansion of the range of things that can happen in museums and museum-like settings' (ibid., p.214). While Clifford's discussions are set in the

[10] See, for example, the research project and publication, *Art-Who Needs It? The Audience for Community Arts*, by Lewis, Morley and Southwood (1986) which attempted to find out whether the GLC's 1981-1986 programme to reach those groups traditionally excluded from the arts actually succeeded in broadening the base of arts provision in London. The authors concluded that in order for the arts and leisure activities to attract new audiences outside of the traditional white, middle class, highly educated elite public who traditionally benefited the public spending on the arts, councils needed to utilise and experiment with their existing cultural assets more imaginatively, by, for instance, holding town or city wide festivals, tempered with a marketing approach.

[11] *Moving Here, 200 Years of Migration to England*, <http://www.movinghere.org.uk>.

basement of the Portland Museum of Art, Oregon, where curators meet with Tlingit tribal elders to discuss the Northwest Coast Indian Collection, the ensuing interaction that focused on custodial responsibilities to the clan's cultural heritage has implications for museum collections around the world. Clifford argues, 'the objects of the Rasmussen collection, however fairly or freely bought and sold, could never be entirely possessed by the museum. They were sites of a historical negotiation, occasions for ongoing contact' (ibid., p.194). Clifford's contact zones represent democratic spaces which draw attention to custodial responsibilities and relationships of reciprocity between the museum and the communities they serve, and where the community-driven agendas and potential resources are distributed rather than fixed in the hierarchical structures of the conventional museum. While most collections, held in custodial responsibility by museums on behalf of the community, would not be as spiritually imbued as those negotiated by the Tlingit tribal objects from the Northwest Coast Indian Collection in the Portland Museum of Art, all objects that are owned by the museum are in fact owned by the community and may be understood as sites of crossing.

Thinking about the museum as community clearly demands a radical adjustment to traditional museum practice. According to Bennett:

> Much of the language of community might imply a critique of the more abstract relationships of government or of a state. What stands behind the ecomuseum are the activities of government which, in establishing such museum and training their staff, developing new principles for the exhibition of cultural materials and a host of related tinkering with practical arrangements […] equips it to be able to develop itself as a community. (Bennett 1998, p.202)

When community action is amplified in the ways suggested by the ecomuseum, the new museologists and Clifford's contact zones, it could serve to reverse the traditional role of curator and visitor, transferring agency from the museum back into the community, and setting up new synergies of co-production. While not actually redistributing the artefacts themselves, thinking about the VM as a contact zone suggests that these kinds of co-coordinated, online activities could offer new opportunities to share the knowledge, histories, and interpretations of objects. The ubiquitous electronic networks connect centre to periphery even when the distance is only a bus ride away, enabling opportunities for discourse and essentially replacing the producer/consumer model of the museum/visitor relationship with new co-productions of resources driven horizontally across the networks. The innovative nature of these networks, however, may be deceptive, especially when they serve to perpetuate the legacy of the past. When VMs simply replicate entrenched ideologies, the technological innovation is predicated on archaic principles, and their emancipatory potential is forfeited. Employing new media in the museum, therefore, must be seen as complex and ambiguous, and in describing these enabling technologies, V-Must identifies both those VMs that illustrate the emancipation of the rigid taxonomies of the traditional museum, as well as those that simply enable the museum to re-enact out-dated positions. The role of V-Must is not to critique either of these approaches, rather to identify and map out the different narratives the VMs encapsulate.

To conclude with Clifford's utopian vision of the museum as contact zone, he argues that:

> within broad limits, a museum can accommodate different systems of accumulation and circulation, secrecy and communication, aesthetic, spiritual, and economic value. How its "public" or "community" is defined, what individual, group, vision, or ideology it celebrates, how it interprets the phenomena it presents, how long it remains in place, how rapidly it changes – all these are negotiable. (Clifford 1997, pp.217-218)

V-Must's arguments for a VM are speculative and not prescriptive; we can only retrospectively describe the dynamics and ideologies of this space that is perceived. We would of course prescribe an uncompromisingly democratic space that encompasses plurality and embraces cultural diversity if we perceived our role to influence things in any way, we still wouldn't ignore the reality where some VMs still clearly remain locked in fossilised legacies that still do not open their doors to everyone. To end on a more optimistic note, however, we argue that the trend is towards a greater openness, and without wishing to sound visionary or futuristic, we sense that museums, in a post Web 2.0 world are working towards a more open approach. In some cases they are actively integrating the micro-histories of their public into the meta-narratives of national histories, and accommodating broader audiences who come to the museum with their own agendas. Terms like the 'digital divide' have been replaced with ideas of social inclusion, and agendas developed to enhance cultural diversity and pluralism are gathering ground. To evaluate the role of VMs in this progression, the following examples represent how some experiences reiterate the legacy ideologies of the traditional museum, while at the same time other

platforms have opened up unique opportunities for open discussion and an opportunity to welcome all voices from communities across all contact zones.

The following section discusses three scenarios of the VM. In their own way, each describes the relationship between the museum and the visitors; each one highlights the progression from physical visit to virtual and the different kinds of opportunities that they afford the user/visitor. Whether they represent the truly democratising approach that the New Museologist advocates or not, these kinds of scenarios do offer the potential for re-shuffling the cards from a traditional museum approach to one that is more in tandem with the Web 2.0 world.

Yoga: The Art of Transformation: before the visit, the VM and the marketing scenario

Visitors to the exhibition *Yoga: The Art of Transformation* that opened at the Arthur M. Sackler Gallery in Washington October 19th, 2013 may well have heard of it months in advance. Launching the exhibition across the Smithsonian's blog, via their E-newsletter, Twitter, YouTube Channel and over Facebook, the Museum had ample time to spread the Yoga vibes to potential audiences. Introducing the visitor to the exhibition and inviting them to 'use' the exhibition in novel ways served not only to introduce the visitors to the exhibition, but also to become directly involved in promoting it. Crowdsourcing the museum audience in this way amplified the exhibition's message and served to draw in potential visitors to the exhibition. The term 'crowd sourcing' is used in connection with a whole range of online activities, and generally describes the ways in which the public - that is you and I - are harnessed by others to do their work for them. This is a dynamic give and take relationship where it seems that some are giving while others take. Crowd sourcing sets up all sorts of novel power relationships: in some scenarios individuals act out of pure altruism (doubtful); with others it seems that there is one side gaining more than the others (usually by those who are doing the outsourcing); in yet other scenarios there are those who are prepared to be resources for their own-altruistic pleasure. As they say, 'it's complicated!' (Hazan 2011) Jeff Howe[12] was the first person to use the term when he described the concept of crowdsourcing in Wired Magazine as how smart people find ways to tap the latent talent of the crowd.

The cozy invitation to 'Get Involved with Yoga' included invitations to the numerous events, symposia, workshops, gallery tours, performances, demonstrations, festivals and family activities planned around the exhibition – fairly standard procedure for museums these days. But what made this call to action outstanding was the warm embrace of not only the museum visitors but the entire yoga community. 'Thank you so much for your interest in Yoga: The Art of Transformation, the world's first exhibition of yogic art' the museum intoned, 'we are honored that so many people in the community have contacted the Freer|Sackler, wanting to get involved'. In addition to the social network promotion, the curatoral staff distributed videos, slideshows and even printable posters to pin up in their own yoga studios, in coffee shops, schools, or other gathering places so that the community could share a little of the magic of 'the world's first exhibition on yogic art' at the Smithsonian in Washington.

[12] http://www.wired.com/wired/archive/14.06/crowds.html



Screenshot: The Art of Transformation website, Sackler Gallery, Smithsonian Museum, Washington, <http://www.asia.si.edu/support/yoga/default.asp>

But the call to action and the community embrace didn't end here, the museum also outlined a number of ways in which individuals and organizations could participate, including sponsoring a program, promoting the exhibition online, and dedicating a yoga class to support the show. Visitors were invited to look over the opportunities offered, complete the online form to let them know how they would like to become part of yoga history. Here was an invitation to crowdfund the exhibition that included works of Indian art, including temple sculptures of tantric goddesses, film clips of early twentieth-century yogis, and colorful manuscripts of ascetics journeying across the countryside.

In order to assist the Museum to 'trace yoga's central tenets and profound meanings over 2,000 years' the museum launched their crowdfunding campaign, aimed at the yoga crowd, asking them to 'donate now' to enrich their understanding of yoga and Indian culture. For a mere $25 visitors could support 'Serenity', for $65 they could 'Help create tranquil galleries'. The 'Power' donation of $65 'brings yoga classes to the museum'. For $150 you receive your share of 'Bliss by sharing of concerts, workshops & festivals'. $500, that is if you are feeling particular enlightened, brings you 'Transformation' which 'Turns knowledge into books'. Finally opportunities to donate $1,000 to choose 'Flight' promises to Transport yoginis across the world.

At the time of writing this paper (one week before the opening), the Museum had already raised $176,415 towards their goal of $200,000 and if they reached all their other crowdsourced goals in the same way, they were well positioned to open up their doors to welcome their very first visitor.

Through a simple search on Kickstarter,[13] there are currently 260 Museum projects, including some that are already fully funded including *CUBIST*, a desktop game that builds a 'grand and inspiring' new Modern Art Museum[14] including its interior sculptures or "installations." Out of cubes, or more precisely, dice! CUBIST was 160% funded, with $16,007 pledged with 236 backers.

Collection Wall: the electronic delivery that accompanies the visitor during the visit

While some museums offer the traditional small screen gallery guide, either on mobile devices offered by the museum for the duration of the visit or over BYODs, which stands for "Bring Your Own Device", content follows the visitor as he or she moves around the gallery or exhibition. Other museums use different sizes of screens or responsive tables located on the gallery floor that call up collections or present a virtual tour at the visitor's call. The second case study briefly describes an engaging scenario of electronic delivery, opened on January 21, 2013 that accompanies the visitor during the visit to Gallery One, the 13,000-square-foot atrium at the Cleveland Museum of Art (CMA). *Collection Wall*[15] vividly visualizes all the works currently on view in CMA's permanent collection galleries, as well as those that are held behind the scenes in storage. All together over 3,500 works of art are on display which adds up to more than 23 million pixels to play with!

According to the Museum's website:[16]

> The Collection Wall is the largest multi-touch screen in the United States—a 40-foot, interactive, microtile wall featuring over 3500 works of art from the permanent collection, most of which are on view in the galleries. The display changes every 40 seconds, grouping works by theme and type, such as time period, materials and techniques, as well as 32 curated views of the collection.

Pitched essentially as an orientation experience, the interface facilitates discovery and dialogue both with the collections and with other visitors, allowing them to download existing tours, or create their own tours which they can then take out into the galleries on iPads. The *Collection Wall* in this way serves to personalise the visit, and allows visitors to appropriate the objects and works of art; connecting with objects in ways that are bespoke and therefor meaningful for them.



Cleveland Museum of Art, (CMA) Collection Wall

For those who can't make up their mind they can spend a little time at the wall and wait for it to change its curated view. To discover one of the groupings of objects from the collection, organized around curated themes like "Love and Lust," "Funerary Art," and "Dance and Music." the collections have also

[13] http://www.kickstarter.com/projects/search?utf8=%E2%9C%93&term=museum
[14] http://www.kickstarter.com/projects/167427101/cubist-build-an-inspiring-new-art-museumwith-dice?ref=search
[15] http://www.clevelandart.org/gallery-one/collection-wall
[16] http://www.clevelandart.org/gallery-one/collection-wall

been organised by medium or geographical region, all called up on the fly from the CMA's digital asset management system. The museum describes this experience as a:

> huge interactive tool [that] allows visitors to see the permanent collection as a living organism, changing depending on the prism through which you view it. The Collection Wall further functions as a giant group and individual touchscreen interactive, and allows visitors to touch the objects represented on the wall to make discoveries. Visitors follow their curiosity through a visual interface that links each artwork to a series of associated artworks, giving visitors the opportunity to browse and explore relationships from object to object. (Alexander, Barton, and Goeser 2013)

Having made their selection, visitors save their favorites onto their iPad by placing their device on one of eight docking stations, which identify an iPad by detecting an RFID chip on the back of its case. Visitors can download it for free to their iPads, or pre-loaded iPad 4s are available to rent on location for a nominal fee of five dollars. In addition, the visitor's favoriting, and sharing activity explains the museum, creates metrics that enable museum staff to understand what artworks visitors are engaging with, creating a feedback loop with the museum.

Once out and about in the gallery with their bespoke tour uploaded to their personal device, visitors can use one of the three additional functions as described by staff at the CMA:

> The "Near You Now" function allows visitors to browse and find digital interpretations of works of art they like based on proximity. Content is designed in short segments of audio and video, allowing visitors to choose what they want rather than committing to a long, linear narrative. Visitors can hear from curators, educators, and community members to discover the continuing traditions that bring art to life.
>
> The "Tours" function allows visitors to have a more structured experience in the galleries, taking a tour curated for the block of time they have available. They can walk through the galleries with CMA's director to discover his favorites, or they can follow a theme that carves a focused path through the museum's galleries. The two hundred most recently saved Visitor-created tours are also available.
>
> The "Scan" function uses image recognition to allow visitors to scan two-dimensional art objects to trigger texts or videos to pop up on the iPad screen. The immediate delivery of this additional interpretive content enables visitors to delve more deeply into the app to learn more about a work of art.

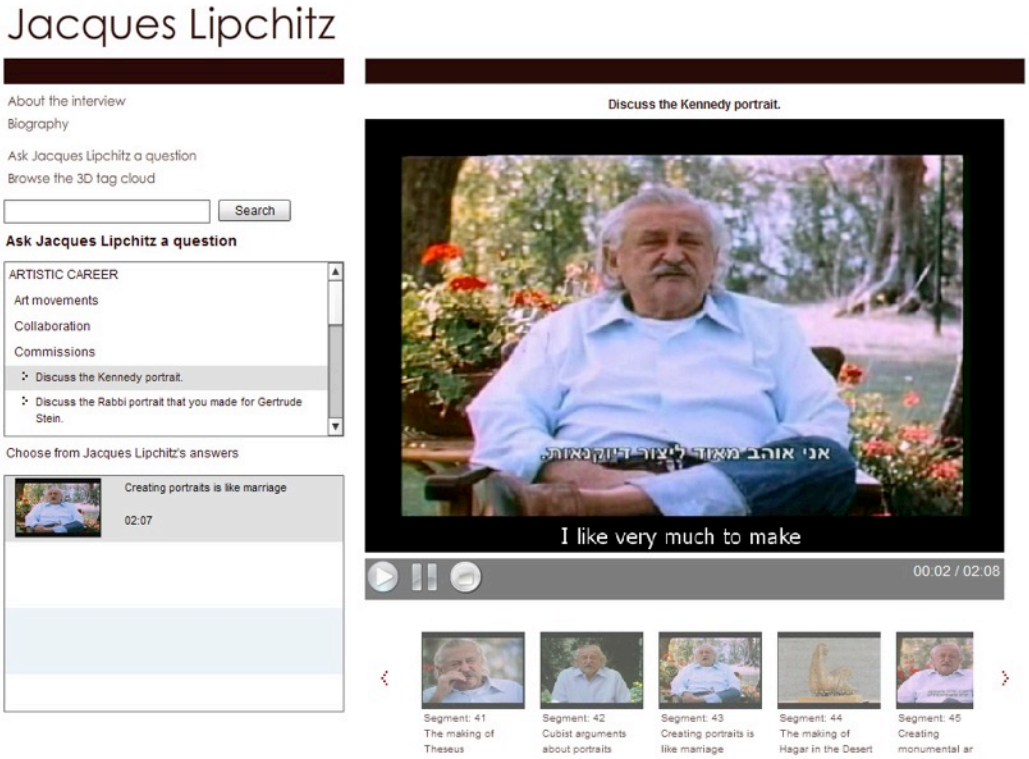
In this scenario it is difficult to separate whether it is the screen that is responding to the visitor or the visitor to the screen, but what is clear here, is the symbiotic accommodation of the museum to its audience, and the potential for optimising the visit, and, in doing so empowering the visitor to take on an active role that is tailor made to him or her during the gallery tour.

Ask Jacques Lipchitz a Question: the VM as encountered after the visit that serves to augment and enhance the experience and maintain the connection between the Museum and the visitor

The third VM scenario, *Ask Jacques Lipchitz a Question*,[17] authored by the Israel Museum, Jerusalem[18] and developed by STARC, at the Cyprus Institute is an experience that can be enjoyed after the visit. Recalling the Lipschitz works explored during the Museum visit, the platform serves, in a novel way, to augment and enhance the experience and maintain the connection between the Museum and the visitor through opening up a unique opportunity to 'meet the artist' and 'hear his voice' – even after his own death.

[17] http://www.imj.org.il/lipchitz

[18] Digitization for the IMJ presentation of the conversations: Mr. Hanno D. Mott, New York for the family of Jacques Lipchitz. Interviewer and video producer: Bruce W. Bassett, 1971-2002. With the assistance of the Computer and Information Systems Department © The Israel Museum, Jerusalem.



Screenshot: Ask Jacques Lipchitz a Question, The Israel Museum, Jerusalem

Jacques Lipchitz (August 22nd, 1891 - May 16th, 1973) was interviewed by Bruce Bassett during the summers of 1970 to 1972, at Villa Bosio in Pieve di Camaiore, Italy. Over this web-based platform, visitors are invited to hold a conversation with Jacques Lipchitz by asking questions, searching through his ideas, or by following term tags and then listening to his video clips as he answers your specific query.

A pioneering artist of the 20th century, Chaim Jacob Lipchitz (1891-1973) was born in Druskininkai, Lithuania. In 1909 Lipchitz moved to Paris to study at the École des Beaux-Arts and at the Académie Julian, and also attended drawing classes at the Académie Colarossi. Whilst in Paris he became acquainted with Pablo Picasso, Amedeo Modigliani, and Juan Gris. Lipchitz frequented museums and became deeply interested in ancient and non-western art, and began collecting various artifacts. Following the Nazi occupation of Paris, Lipchitz escaped to New York, where he continued to sculpt until his death. He is buried in Har Hamenuhot, Jerusalem.

In addition to the project's online presence, the platform is on permanent exhibition in the Lipschitz Gallery in the Israel Museum, Jerusalem and Lipchitz Exhibition at Museum of Palazzo Pretorio, Prato, Italy.[19]

Maintaining the connection between the museum and its audience essentially means building a long-term relationship. In this way the VM can provide ideal opportunities to strengthen their links. Whether or not the actual visit inspires visitors to return and learn more, these kinds of platforms create long-term connections and augment the physical visit with echoes of the art and artworks, in this case with the artist's own voice lingering on, even after his own death.

[19] http://www.palazzopretorio.prato.it/pagina2_home.html

Conclusion

One of the main challenges of the VM in the context of new museology - resonating with the debate of the "new archaeology" of the 1970s - is the breaking down of the barriers of agreeable narratives and romantic stories about the past. In the context of archaeological research, researchers are now engaged in "hard core" sciences in order to reveal the past including chemical analyses, physical investigations, statistical methods, etc… The museum still displays artifacts along traditional trajectories – canonical narratives, chronicling the historical past in pre-determined meta-narratives. The VM now offers new opportunities to reset these paradigms.

VM is not (only) about innovative technologies, the main challenge lies in daring to dream, that museums are more than collections of objects, charged with ideology (cultural, elitist, intellectual, richness, prestige, etc…). By the willingness of curators, museum directors or politicians to promote culture, virtual museums can, and should be, true cognitive technologies, platforms of situated learning environments, (cyber) social spaces of interaction, where people meet and learn from each other, through the experience of each other, and through interaction with each other in a new iteration of Clifford's "contact zones". Being digital, VMs can link the Chinese citizen with the Swedish or the South-African with the Russian. In this way, everyone's voice has equal power and intensity, and it's up to the visitor, and perhaps it's their duty, to contribute their own knowledge to the "global" experience of VM.

VM allows almost anything and can digitally "materialize" almost everything. It is up to the curator or perhaps the "digital curator" (a hybrid between a dreamer, a social scientist, an art historian and a computer expert) to explore and fully benefit from the new (social, cognitive and cultural) space. VM is enabling a voyage into a new, and yet unexplored world...

> Space: The final frontier
> These are the voyages of the Starship, Enterprise
> Its 5 year mission
> To explore strange new worlds
> To seek out new life and new civilizations
> To boldly go where no man has gone before
>
> (From the original *Star Trek* science fiction television series. It refers to the mission of the original starship *Enterprise*. The complete introductory sequence, narrated by William Shatner at the beginning of every episode of *Star Trek,* Desilu Productions, and by Paramount Television from 1968-69).

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